

Domain A Measurements

Every number the system reports about a lesson, what each one needs before it can exist, and the five pieces of work that unlock the rest. Teacher only — nothing here measures students.

5 Live today	2 Need only the timetable	15 Need audio	1 Partly built
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Twenty-three measurements, drawn from *Domain A.xlsx* — a three-band rubric (Emerging / Developing / Competent) over timely start and finish, routines and behaviour, and learning time. Each parameter in that sheet carries its own list of system measures; this is the subset being built.

Two pairs that are not the same number

The rubric is strict about this, and it is the easiest thing to get wrong.

R1 Arrival — when the teacher walked in R7 Lesson start — when the first learning task was set <i>A teacher can be in the room ten minutes early and still start late. The rubric grades R7.</i>	R3 Departure — when the teacher walked out R9 Lesson end — when teaching stopped <i>A lesson can end ten minutes before anybody leaves the room.</i>
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Facts someone has to type in

None of these are detected. Seven measurements are impossible without them, and they are the cheapest work in this document — a database column and a form field, no machine learning at all.

ID	INPUT	UNLOCKS
P1	Scheduled start time	R2, R8, R11, R12
P2	Scheduled end time	R4, R5, R11, R12, R16
P3	Subject and year group	Context for R19, R20
P4	Room type Classroom, lab, PE, library, practical	The reliability caveat — movement and noise norms differ
P5	Does the same class have a following period?	R14

Group A — Attendance

ID	THE NUMBER	SENSOR	HOW	STATUS
R1	Arrival time First moment the teacher is in the room	CAMERA	Start of the first presence interval	● Live
R2	Arrival against the bell Minutes early or late	CAMERA + P1	R1 – P1	● Needs timetable
R3	Departure time Last moment she is in the room	CAMERA	End of the last presence interval, or the final door exit	● Live
R4	Departure against the bell Minutes early or late	CAMERA + P2	R3 – P2	● Needs timetable
R5	Time in the room Total, and as a share of the scheduled period	CAMERA + P1/P2	Sum of presence intervals	● Live ● share needs timetable
R6	Mid-lesson absences How many times she left, and for how long	CAMERA	Gaps between presence intervals, matched to door crossings	● Live

Group B — The lesson

ID	THE NUMBER	SENSOR	HOW	STATUS
R7	Lesson start When the first learning task was set	MIC	First teacher utterance that launches a task — starter, do-now, retrieval or discussion	☐ Not built
R8	Start delay Minutes from the bell to the first task	MIC + P1	R7 – P1	☐ Not built
R9	Lesson end When teaching stopped	CAMERA + MIC	Last teaching utterance, corroborated by the teacher leaving the board or the room	☐ Not built
R10	Lesson duration	CAMERA + MIC	R9 – R7	☐ Not built
R11	Did the lesson fit the period?	CAMERA + MIC + P1/P2	Whether [R7, R9] sits inside [P1, P2]	☐ Not built
R12	Overrun or underrun Minutes past the bell, or minutes of the period left unused	CAMERA + MIC + P2	R9 – P2, in either direction	☐ Not built

R8 is the number that grades punctuality. Everything else in this group is reported alongside it so the figure can be read honestly — a lesson that started on time and ended fifteen minutes early is not the same as one that ran the full period.

Group C — How the lesson ended

ID	THE NUMBER	SENSOR	HOW	STATUS
R13	Closure, and its type Review, reflection, exit question, summary, or none	MIC	Teacher utterances near the end labelled as closing	☐ Not built
R14	Continuation Did she say the lesson carries on next time	MIC + P5	An explicit statement that the topic continues	☐ Not built

ID	THE NUMBER	SENSOR	HOW	STATUS
R15	Homework set Whether homework was given, and when	MIC	Teacher utterance assigning work beyond the lesson	○ Not built
R16	Pack-up instruction When the class was told to pack up, against the bell	MIC + P2	First pack-up instruction, minus P2	○ Not built

R13 and R14 answer the same question from two directions. A lesson can end properly with a summary, end properly by being explicitly carried over, or just stop. Those are three different outcomes and the report names which.

Group D — Voice and delivery

ID	THE NUMBER	SENSOR	HOW	STATUS
R17	Raised-voice events Count, and rate per ten minutes	MIC	Loudness of the teacher's own speech against her own rolling baseline, sustained past a minimum duration	○ Not built
R18	Attention requests How many times she called for the class's attention	MIC	Teacher utterances flagged as an attention cue	○ Not built
R19	Off-lesson drift How many episodes, and total minutes	MIC	Runs of teacher speech labelled as unrelated to the lesson	○ Not built
R20	Questions asked Count and rate, split into questions put to the class and rhetorical check-ins	MIC	Teacher utterances labelled as asking	○ Not built
R21	Languages used Which, how many, the share of speech in each, and switches per minute	MIC	Per-utterance language, normalised before counting	○ Not built

R17 is loudness, not words. Transcription returns no volume at all. This is a separate pass over the audio waveform, and it is the only measurement here that does not read the transcript.

R20 must separate real questions from filler. “ठीक है?”, “समझे?” and “right?” are punctuation, not questions. Counting question marks overstates this number several times over.

R21 is not in the rubric. It is an addition, and a useful one in a classroom that code-switches. Report the set of languages, the share of each, and the switch rate — the last is the interesting one.

Group E — Trust

ID	THE NUMBER	SENSOR	HOW	STATUS
R22	Observation coverage How much of the lesson the system could see and hear, and how confident it is	CAMERA + MIC	Coverage, continuity and detection confidence, reported per sensor	● Live for video
R23	Not Observed The system says so rather than guessing	—	Any measurement below its evidence threshold reports Not Observed	● Partly built

What has to be built

Five pieces of work, in the order that returns the most for the least. Step 1 is a database column and a form field. Step 2 is the large one — fifteen measurements sit behind it.

- 1 Timetable fields**
— scheduled start and end on every lesson
R2 · R4 · R5 · R8 · R11 · R12 · R16
- 2 Audio capture and transcription**
— the teacher's voice separated from everyone else's
R7 · R9 · and all of Groups C and D

3 Loudness pass

— over the audio waveform

R17

4 Utterance labelling

— what each teacher utterance was doing

R13 · R14 · R15 · R16 · R18 · R19 · R20

5 Language detection and normalisation

R21

Known measurement risks

Both found in testing. Both affect Group D.

- **Student speech gets absorbed into the teacher's turns.** A single room microphone cannot reliably separate one teacher from thirty students, so anything measured as “the teacher's speech” runs high. A clip-on microphone for the teacher plus a room microphone removes this entirely, and is the recommended recording setup.
- **The same word appears in two scripts.** In code-switched audio an English term can be transcribed in Latin or in Devanagari within one lesson. Text has to be normalised before anything counts occurrences, or R19, R20 and R21 all undercount.

Deliberately out of scope

Not measured, and not to be inferred. Where the rubric grades one of these, the system reports *Not Observed* rather than a number it cannot stand behind.

- **Anything about students** — whether they were working, out of their seats, packing up early, or settled. The system does not detect students.
- **Transitions between activities** — count, length or cost.
- **Settle-after-cue time** — it requires knowing whether the class went quiet *and* still, which needs the students.
- **Learning-time share** — a percentage of the period spent learning cannot be produced honestly without the above.

- **Off-task noise** — meaningless without knowing what the class was meant to be doing.
- **Lesson-plan phase context** — no lesson plan is required as an input.

Rules that apply to every number

From the rubric's own closing note.

1. Never judge from a single moment — a signal must persist to count.
2. Anchor start and finish to the timetable, never to the length of the recording.
3. Where the observation is insufficient, state **Not Observed**.
4. Labs, PE, library and practical lessons have different movement and noise norms; lessons with no audible bell, and double periods, break the timing assumptions. In those cases the reading is declared unreliable.

Source: *Domain A.xlsx*, sheet “Domain A”. Maintained in the repository as `docs/teacher-measurements.md`.